

Bioreactor Output : Algae Biomass Production and CO₂ Capture

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Expected Daily Dry Chlorella Vulgaris Biomass Production

Goal: To find out how many kilograms of dry Chlorella Vulgaris biomass the bioreactor needs to produce daily to successfully remove targeted amounts of Nitrogen and Phosphorus from the wastewater.

- Daily wastewater flow (Q): 2,400 L/day
- Assumed Influent Phosphorus Concentration (P_{in}): 7 mg/L
- Assumed Influent Nitrogen Concentration (N_{in}): 40 mg/L

The standard algae Composition Values (for Chlorella Vulgaris):

- Nitrogen (N) Content in Dry Algae : 8%
- Phosphorus (P) content in Dry Algae : 1%

- Step 1: Calculating the daily mass of Phosphorus (P) to be removed

This calculates the total mass of phosphorus, in milligrams, that needs to be extracted from the wastewater daily based on my targeted removal efficiency.

$$\begin{aligned} P_{\text{removed}} (\text{mg P/day}) &= P_{\text{in}} (\text{mg/L}) \times Q (\text{L/day}) \times \left[\frac{\text{Target Removal Efficiency}}{100} \right] \\ &= (7 \text{ mg/L}) \times (2400 \text{ L/day}) \times \frac{95}{100} \\ &= 15,960 \text{ mg P/day} // \end{aligned}$$

- Step 2: Calculate the daily Mass of Nitrogen (N) to be removed

Similar to step 1 this step calculates the total mass of nitrogen, in milligrams, that needs to be removed daily

$$N_{\text{removed}} = 40 \text{ mg/L} \times 2400 \text{ L/day} \times \frac{95}{100}$$

$$\text{Nitrogen removed} = 91,200 \text{ mg N/day} //$$

- Step 3: Calculating Daily Dry Biomass Required based on Phosphorus Removal

$$\text{Biomass}_P (\text{mg dry algae/day}) = P_{\text{removed}} (\text{mg P/day}) / (\text{P content in dry algae})$$

$$\text{Biomass}_P = 15,960 \text{ mg P/day} / 0.01 (\text{since P content is 1\%})$$

$$\text{Biomass}_P = 1,596,000 \text{ mg dry algae/day}$$

$$\begin{aligned} &\text{Divide } 1.596 \text{ by } 1,000,000 \text{ (1 kg = 1,000,000)} \\ &= \frac{1,596,000}{1,000,000} = 1.596 \text{ kg algae/day} // \end{aligned}$$

- Step 4: Calculate Daily Dry Biomass Required Based on Nitrogen (N) Removal

$$\text{Biomass}_N (\text{mg dry algae/day}) = N_{\text{removed}} / (\text{N content in dry algae})$$

$$\begin{aligned} \text{Biomass}_N &= 91,200 \text{ mg N/day} / 0.08 \\ &= 1,140,000 \text{ mg dry algae/day} \end{aligned}$$

Converted to kg:

$$1,140,000 \text{ mg} / 1,000,000$$

$$\text{Biomass}_N = 1.140 \text{ kg dry algae/day}$$

Step 5: Determining Final Expected Daily Dry Chlorella vulgaris Biomass Production

$$\begin{aligned} \text{Biomass}_P &= 1.596 \text{ kg/day} \\ \text{Biomass}_N &= 1.140 \text{ kg/day} \end{aligned} \quad \begin{array}{l} \nearrow \text{choosing the} \\ \nearrow \text{bigger value!} \end{array}$$

Final Result for daily dry Chlorella vulgaris Biomass Production:

$$\Rightarrow 1.596 \text{ kg dry algae/day}$$

What this means:

Based on the calculated nutrient removal requirements (from an influent of 7mg/L P and 40mg/L N with 95% removal efficiency) within the 15,600 Liter bioreactor, the system is expected to produce approximately 1.6 kg of dry Chlorella Vulgaris algae biomass per day.

Quantifying CO₂ Capture

Objective: To calculate how much CO₂ the bioreactor is estimated to capture daily, based on the algae biomass produced.

Assumption: Carbon content in dry algae: Approx. 50% (standard average value)

Step 1: Calculate the mass of carbon contained in the algae biomass

This step the amount of pure carbon locked up in dry algae biomass produced each day.

$$\begin{aligned} \text{Mass of Carbon (kg C/day)} &= \text{Expected daily dry algae biomass (kg dry algae/day)} \times \text{Carbon content in dry algae in decimal} \\ &= 1.596 \text{ kg dry algae/day} \times 0.50 \text{ mass of carbon} = 0.798 \text{ kg C/day} // \end{aligned}$$

Step 2: Convert Mass of Carbon to Mass of CO₂ captured

This step uses the molecular weights of Carbon (C) and carbon dioxide (CO₂).

Molecular Weight:

$$\text{Carbon (C)} \approx 12 \text{ g/mol}$$

$$\text{Carbon dioxide} \approx 44 \text{ g/mol}$$

$$(12 + 2(16)) = 44 \text{ g/mol}$$

This means: every 12 units of carbon → come from 44 units of CO₂

$$\text{Formula: Mass of CO}_2 \text{ captured} = \text{Mass of Carbon (kg C/day)} \times (44/12)$$

(kg CO₂/day)

$$= 0.798 \times (44/12) = 2.96 \text{ kg CO}_2/\text{day}$$